



FDA CDER analysis of non-human primate (NHP) data in monoclonal antibody (mAb) development to support the streamlining of nonclinical safety studies

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ABSTRACT

A retrospective analysis of nonclinical study reports submitted in regulatory packages of FDA-approved mAbs, excluding antibody–drug conjugates, was conducted. This analysis was limited to the period between January 2020 and April 2025. For highly target-specific mAbs, the incremental contribution of long-term animal studies to regulatory safety assessment was evaluated. FDA review documents were examined to assess study duration, species selection, and the regulatory relevance of toxicological findings to determine whether long-term *in vivo* studies submitted in Biologics License Applications (BLAs) identified new safety signals beyond those seen in shorter-term studies.

Among the mAbs evaluated, long-term NHP toxicity studies were included in more than half of the development programs but rarely altered regulatory conclusions regarding human risk. Additional findings were predominantly attributable to known pharmacology or species-specific immunogenicity and generally did not affect no-observed-adverse-effect-level (NOAEL) determination, clinical trial design, or labeling. No long-term study in this analysis identified unanticipated off-target toxicities that fundamentally altered the overall safety assessment. These results are consistent with principles articulated in the FDA's 2025 Roadmap for Reducing Animal Testing in Preclinical Safety Studies and provide descriptive information relevant to recent draft FDA guidance on streamlined nonclinical safety studies for monospecific mAbs.

1. Introduction

On April 10, 2025, the U.S. Food and Drug Administration (FDA) released the roadmap for Reducing Animal Testing in Preclinical Safety Studies (U.S. Food and Drug Administration, 2025a). Within its proposed implementation actions over the next three years, mAbs were identified as a promising area for reducing animal use because the nature of these therapeutics. The roadmap mentions anti-drug antibody (ADA) and the immunogenicity of mAbs as factors which can alter drug exposure and confound toxicity interpretation in long-term (≥ 26 weeks) NHP studies. In addition, it is important to note that mAbs have human-specific targets, restricted tissue distribution, lack of hepatic biotransformation, and reduced potential for off-target toxicities. Accumulated evidence demonstrates the limited predictive value of traditional animal models for safety endpoints, and support the

statements in the roadmap. The roadmap emphasizes that study design and duration should be guided by scientific rationale rather than historical convention and encourages the integration of new approach methodologies (NAMs). Among the options outlined for the next three years, the roadmap proposes reducing routine ≥ 26 weeks NHP repeat-dose toxicity studies for mAbs that do not show concerning findings in shorter-duration studies (4- to 13-week studies), when supported by complementary NAMs data. The roadmap also indicates that FDA plans to develop or update guidance to clarify how such NAMs can be used to support reduced animal number, species and duration in nonclinical development programs.

Building on the roadmap, in December 2025 the FDA published a draft guidance describing FDA's current thinking on the nonclinical safety evaluation of mAbs that recognize a single target, referred to as monospecific antibodies (U.S. Food and Drug Administration, 2025b).

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Overall, the guidance provides recommendations for streamlined approaches to assess long-term safety of monospecific antibodies and outlines a framework to reduce reliance on long-term NHP studies, consistent with the roadmap and FDA's commitment to applying the principles of replacement, reduction, and refinement (3Rs). Prior to the release of recent FDA draft guidance, nonclinical safety evaluation of mAbs relied on the International Council for Harmonisation of Technical Requirements of Pharmaceuticals for Human Use (ICH) S6 guideline for biotechnology-derived pharmaceuticals and its revision, S6 (R1), which established the principles for the nonclinical safety assessment of biopharmaceuticals. S6 (R1) emphasizes the selection of pharmacologically relevant species, allows the use of a single relevant species in long-term toxicity studies when scientifically justified, and encourages tailoring study design and duration to the product's mechanism of action and intended clinical use ([International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use, 2011](#)). More recently, the European Medicines Agency (EMA) issued the Reflection Paper on NHPs in Safety Testing of Human Medicinal Products and Opportunities for 3Rs Implementation, emphasizing that the use of NHPs in non-clinical safety testing must be scientifically justified. The paper also discusses the potential to waive long-term toxicity studies, and encourages the application of alternative testing strategies to better predict toxicity outcomes ([European Medicines Agency, 2025](#)).

To support the development of this draft guidance, we in the Center for Drug Evaluation and Research (CDER), Office of New Drugs (OND), Pharmacology/Toxicology (PTOX) conducted a retrospective analysis of nonclinical safety data submitted in support of mAbs development. This analysis was conducted in the context of the 3–5-year implementation plan of the FDA NAMs Roadmap, which includes reducing routine ≥ 26 weeks NHP toxicology testing for mAbs. Specifically, we evaluated NHP repeat-dose toxicity studies included in regulatory packages submitted to CDER and to support the safety assessment of mAb therapeutics. We focused our analysis on determining whether there were any long-term NHP studies identified toxicities which were not observed in shorter-duration studies. The analysis further examined whether findings from long-term studies influenced regulatory decision-making or contributed meaningfully to human risk assessment.

Accordingly, this manuscript provides an evidence-based assessment of the scientific value of long-term NHP repeat-dose toxicity studies for mAb therapeutics, based on nonclinical data submitted in drug applications to CDER and FDA regulatory review experience. The analysis encompasses mAb therapeutics, including monospecific antibodies, multi-specific antibodies, and antibody combination products, and is intended to inform development of the FDA draft guidance and contributes to implementation activities described in the FDA NAMs Roadmap.

2. Methods

2.1. Dataset source

This analysis was restricted to mAbs that were previously approved by FDA, since the drugs under investigation may lack a complete safety package at the time of review. Since 2008, a total of 178 Biologics License Applications (BLAs) have been approved by the FDA. Inclusion of older approvals could introduce variability because some earlier development programs relied on animal studies conducted as early as the 1990s. These studies were performed under scientific and regulatory paradigms that may not reflect current development practices. To ensure consistency and relevance to the contemporary regulatory and scientific standards, we limited our analysis to 76 BLAs approved by the U.S. Food and Drug Administration (FDA) within the most recent five-year period (January 1, 2020 to April 10, 2025). Although the draft guidance focuses on monospecific antibodies, multi-specific and cocktail mAb therapeutics were included here to contextualize broader CDER review experience. Antibody-drug conjugates (ADCs) were excluded from this

analysis because their pharmacological and toxicological profiles are substantially impacted by the conjugated cytotoxic payloads and linker technologies. These components introduce distinct mechanisms of action and off-target toxicities that are not representative of unconjugated mAbs. Including ADCs would have confounded the interpretation of safety data specifically attributable to the antibody backbone. A total of 52 FDA-approved mAb therapeutics were included in the final dataset ([Table 1](#)).

2.2. Data collection

Data for each mAb were obtained from publicly available FDA BLA review documents, with primary emphasis on the Pharmacology/Toxicology sections, accessed through the Drugs@FDA database (www.accessdata.fda.gov/scripts/cder/daf/). When necessary, original nonclinical study reports submitted within the BLA were also reviewed to enable more detailed evaluation of study design and findings. For each product, the following variables were extracted:

1. Species used for toxicity evaluation (e.g., NHPs, rodents, or transgenic animal models);
2. Use of single versus multiple species;
3. Study duration, categorized as short-term (≤ 4 weeks), medium-term (> 4 –13 weeks), or long-term (≥ 26 weeks);
4. Presence of toxicological findings unique in long-term studies;
5. Whether such findings informed clinical risk assessment, trial design, or patient monitoring strategies.

For mAbs with both short-/medium-term and long-term toxicity studies, direct comparisons were made between findings from different study durations to identify toxicity findings that emerged only during extended dosing periods. A schematic overview of the data extraction and evaluation process is provided in [Fig. 1](#).

2.3. Terminology used for the classification of toxicological findings observed in our analysis

To systematically evaluate the incremental value of long-term animal toxicity studies, toxicological findings were categorized into four predefined classes based on their nature, mechanistic plausibility, and relevance to the known or anticipated pharmacology of the mAb. These classifications were applied specifically to findings identified in long-term studies that were not observed in corresponding short- or medium-term studies.

Class 1: No Additional Findings.

No new findings were identified in the long-term studies when compared to short- and medium-term studies. The overall toxicity profile remained consistent across study durations.

Class 2: Pharmacology-Related Findings.

These findings were consistent with the intended pharmacological activity of the test article, including exaggerated or delayed pharmacological effects, target-organ responses, or on-target non-therapeutic toxicities.

Class 3: Immunogenicity-Related Findings.

Findings attributed to immune responses against the test article, such as ADA formation, immune complex deposition, and hypersensitivity reactions.

Class 4: Additional Off-Target Effect Findings.

Unexpected findings not attributable to the intended pharmacology or immunogenicity of the mAb. These effects may represent off-target or previously unrecognized toxicities and were considered potential safety signals that could, in principle, justify longer-duration studies.

2.4. Data analysis

The primary objective of this study was to evaluate the study design,

Table 1

FDA-approved mAbs included in the analysis.

Active Ingredient	Approval Date	Indication
concizumab-mtci	12/20/2024	Hemophilia A and B
cosibelimab-ipdl	12/13/2024	Skin squamous cell carcinoma metastatic
zenocutuzumab-zbco	12/4/2024	Non-small cell lung cancer metastatic; Pancreatic adenocarcinoma metastatic
zanidatamab-hrii	11/20/2024	HER2 positive biliary tract cancer
zolbetuximab-clzb	10/18/2024	Metastatic gastric adenocarcinoma, Adenocarcinoma of the gastroesophageal junction
marstacimab-hncq	10/11/2024	Bleeding prophylaxis, Hemophilia A and B
lebrikizumab-lbkz	9/13/2024	Atopic dermatitis
axatilimab-csfr	8/14/2024	Chronic graft versus host disease
nemolizumab-ilto	8/12/2024	Prurigo nodularis
donanemab-azbt	7/2/2024	Alzheimer's disease
crovalimab-akz	6/20/2024	Paroxysmal nocturnal hemoglobinuria
tarlatamab-dlle	5/16/2024	Small cell lung cancer extensive stage
tislelizumab-jsgr	3/13/2024	Esophageal squamous cell carcinoma metastatic
toripalimab-tpzi	10/27/2023	Nasopharyngeal carcinoma
mirikizumab-mrkz	10/26/2023	Ulcerative colitis
bimekizumab-bkzx	10/17/2023	Plaque psoriasis
pozelimab-bbfg	8/18/2023	CHAPLE syndrome
elranatamab	8/14/2023	Relapsed/refractory multiple myeloma
talquetamab-tgvs	8/9/2023	Relapsed/refractory multiple myeloma
nirsevimab-alip	7/17/2023	Respiratory syncytial virus infection, Lower respiratory tract infection viral
rozanolixumab-noli	6/26/2023	MuSK myasthenia gravis, AChR myasthenia gravis
glofitamab-gxbm	6/15/2023	Diffuse large B-cell lymphoma NOS, Diffuse large B-cell lymphoma recurrent, Diffuse large B-cell lymphoma refractory
epcoritamab-bysp	5/19/2023	Diffuse large B-cell lymphoma recurrent, Diffuse large B-cell lymphoma refractory
retifanlimab-dlwr	3/22/2023	Merkel cell carcinoma
lecanemab-irmb	1/6/2023	Alzheimer's disease
ublituximab-xiyy	12/28/2022	Relapsing multiple sclerosis
mosunetuzumab-axgb	12/22/2022	Follicular lymphoma
teplizumab-mzvw	11/17/2022	Type 1 diabetes mellitus
teclistamab-cqyv	10/25/2022	Relapsed/refractory multiple myeloma
tremelimumab-actl	10/21/2022	Hepatocellular carcinoma non-resectable
spesolimab-sbzo	9/1/2022	Pustular psoriasis
nivolumab and relatlimab-rmbw	3/18/2022	Melanoma, Metastatic melanoma
sutimlimab-jome	2/4/2022	Cold agglutinin disease
faricimab-svoa	1/28/2022	Neovascular age-related macular degeneration; Diabetic macular edema
tebentafusp-tebn	1/25/2022	Uveal melanoma
tralokinumab	12/27/2021	Atopic dermatitis
tezepelumab-ekko	12/17/2021	Asthma
anifrolumab-fnia	7/30/2021	Systemic lupus erythematosus
aducanumab-avwa	6/7/2021	Alzheimer's disease
amivantamab-vmjw	5/21/2021	Non-small cell lung cancer metastatic
dostarlimab-gxly	4/22/2021	Endometrial cancer recurrent, Endometrial cancer
evinacumab-dgnb	2/11/2021	Homozygous familial hypercholesterolemia
ansuvimab-zykl	12/21/2020	Ebola virus infection
margetuximab-cmkb	12/16/2020	HER2 positive breast cancer

Table 1 (continued)

Active Ingredient	Approval Date	Indication
naxitamab-gqgk	11/25/2020	Neuroblastoma
atoltivimab, maftivimab, and odesivimab	10/14/2020	Ebola virus infection
satralizumab	8/14/2020	Neuromyelitis optica spectrum disorder
tafasitamab-cxix	7/31/2020	Diffuse large B-cell lymphoma recurrent, Diffuse large B-cell lymphoma refractory
inebilizumab-cdon	6/11/2020	Neuromyelitis optica spectrum disorder
isatuximab-irfc	3/2/2020	Multiple myeloma
eptinezumab-jjmr	2/21/2020	Migraine prophylaxis
teprotumumab-trbw	1/21/2020	Graves' ophthalmopathy

List of mAbs approved by the FDA between January 1, 2020 and April 10, 2025 and included in the retrospective analysis, with corresponding Biologics License Application (BLA) numbers and approval dates. (Each entry represents a unique FDA-approved mAb therapeutic (n = 52); combination products are counted as a single therapeutic.).

including study duration, species selection and toxicological endpoints measured, as well as regulatory impact of animal toxicity studies conducted for mAbs approved by FDA between January 2020 and April 2025. Particular emphasis was placed on assessing the scientific value of long-term NHP repeat-dose toxicity studies relative to shorter-duration NHP studies. Analyses were conducted across four areas: therapeutic indication, study duration, species selection, and translational relevance of toxicological findings. Both quantitative and qualitative approaches were applied, as described below.

2.4.1. Quantitative analysis

Each mAb was categorized according to therapeutic indication (oncology vs. non-oncology), species used for nonclinical safety assessment (NHPs, rodents, surrogate antibodies, or transgenic models), and study duration. Surrogate antibodies refer to species-reactive antibodies used in animal studies when the clinical antibody does not bind the homologous target in the test species. The frequency and distribution of these study types were analyzed to identify trends in nonclinical safety packages and to assess alignment with existing regulatory guidance, including ICH S6 (R1) and ICH S9 ([International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use, 2011](#); [International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use, 2017](#)).

2.4.2. Qualitative analysis

For mAbs with both short/medium- and long-term toxicity studies, toxicological findings were compared across study durations to determine whether long-term studies yielded novel or clinically meaningful information. Newly identified findings in long-term studies were classified according to the four predefined categories described in Section 2.3.

2.4.3. Regulatory impact assessment

Each toxicological finding identified in the long-term studies was further evaluated for its regulatory significance. Specifically, we assessed whether the finding:

1. Influenced the determination of the NOAEL;
2. Altered clinical trial design, dosing, or patient monitoring strategies;
3. Resulted in labeling statements or post-marketing requirements.

Regulatory review documents were examined to identify cases in which long-term animal studies directly informed FDA recommendations or risk management decisions.

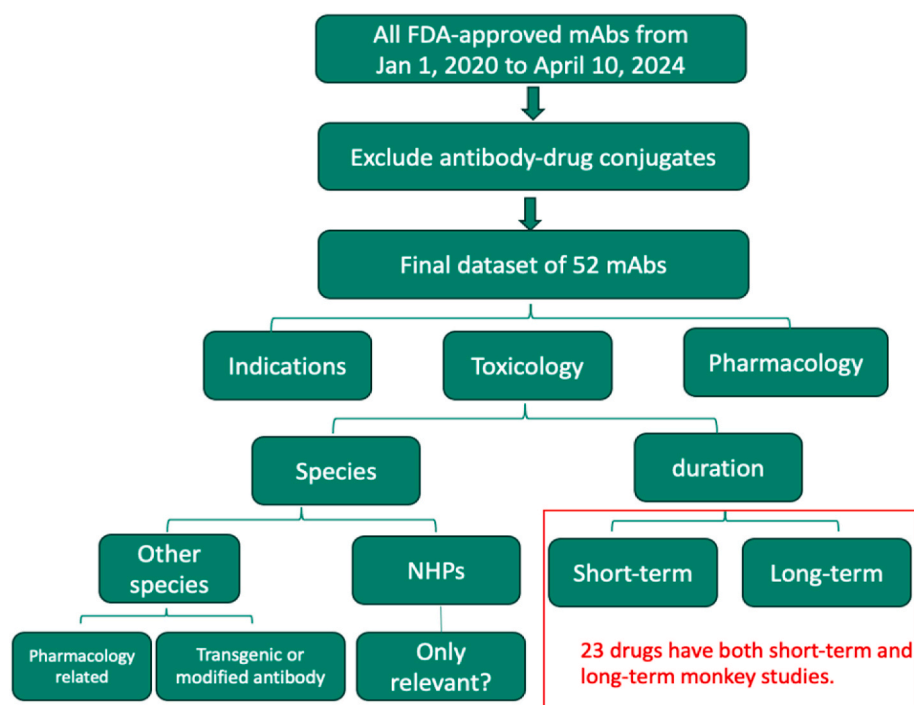


Fig. 1. Workflow of this analysis.

Overview of the data sources, inclusion criteria, and analytical steps used to evaluate nonclinical toxicity study duration, species selection, and regulatory relevance for FDA-approved mAbs.

3. Results

3.1. Study duration and therapeutic context

Among the 52 mAbs approved by the FDA between January 2020 and April 2025, 51 (98.1%) were supported by nonclinical toxicity studies in animals, consistent with expectations under ICH S6 (R1). One notable exception is Tebentafusp, which exclusively recognizes human HLA. Due to absence of pharmacologically relevant animal species to evaluate its toxicity, the entirety of the nonclinical safety evaluation relied on *in vitro* approaches, including cross-reactivity assays and functional assays to assess potential organ-specific toxicities.

Of the 52 mAbs in this analysis, 23 (44.2%) were approved for oncology indications and 29 (55.8%) for non-oncology indications (Fig. 2). Within the non-oncology group, Immunology and Autoimmune Disease accounted for the largest portion (41%), followed by Neurodegenerative Disorders (17%), and Hematologic conditions (17%).

Medium to long-term repeat-dose toxicity studies (≥ 13 weeks) were submitted for 41 of 51 mAbs (80.4%), whereas the remaining 10 (19.6%) relied solely on short-term studies of four weeks or less to

support approval (Fig. 3). Among these 10 mAbs, six were oncology drugs, three were indicated for viral infections, and one was for Type 1 diabetes. Notably, 28 mAbs (54.9%) included long-term repeat-dose toxicity studies.

Study duration was strongly associated with therapeutic indication. Consistent with ICH S9 guidance for oncology therapeutics, streamlined (no long-term studies, only one species) nonclinical packages were more common in cancer drug submissions (International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use, 2017). Of the 23 oncology mAbs, short-to medium-studies were often sufficient to support approval. Only three oncology mAbs (13%), toripalimab, mosunetuzumab, and tremelimumab, included 26-week or longer toxicity studies, due to factors such as combination use, novel targets, or immune-related risk considerations. In contrast, mAbs developed for non-oncology indications were much more likely to include 13-week or longer studies. In the 28 drugs that submitted long-term studies, 26 (92.9%) were non-oncology drugs. Furthermore, 10 drugs had both medium- and long-term studies, and all of these were for non-oncology indications, 13 drugs had both short- and long-term studies, with only two out of these being for oncology indication

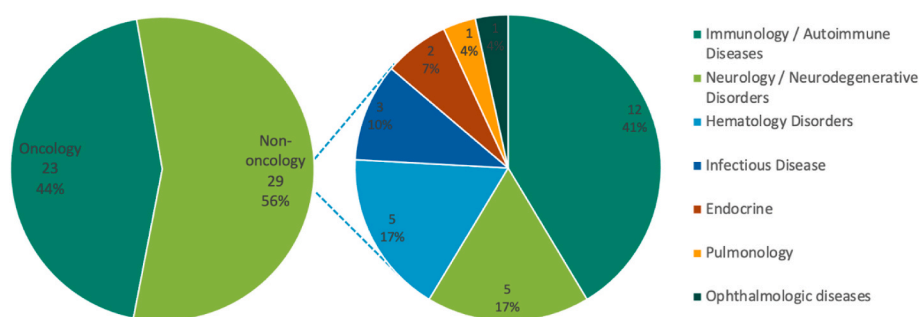


Fig. 2. Drug indication analysis.

Distribution of FDA-approved mAb therapeutics (2020–2025) by therapeutic indication. Oncology and non-oncology categories are shown, with non-oncology indications further classified by disease area.

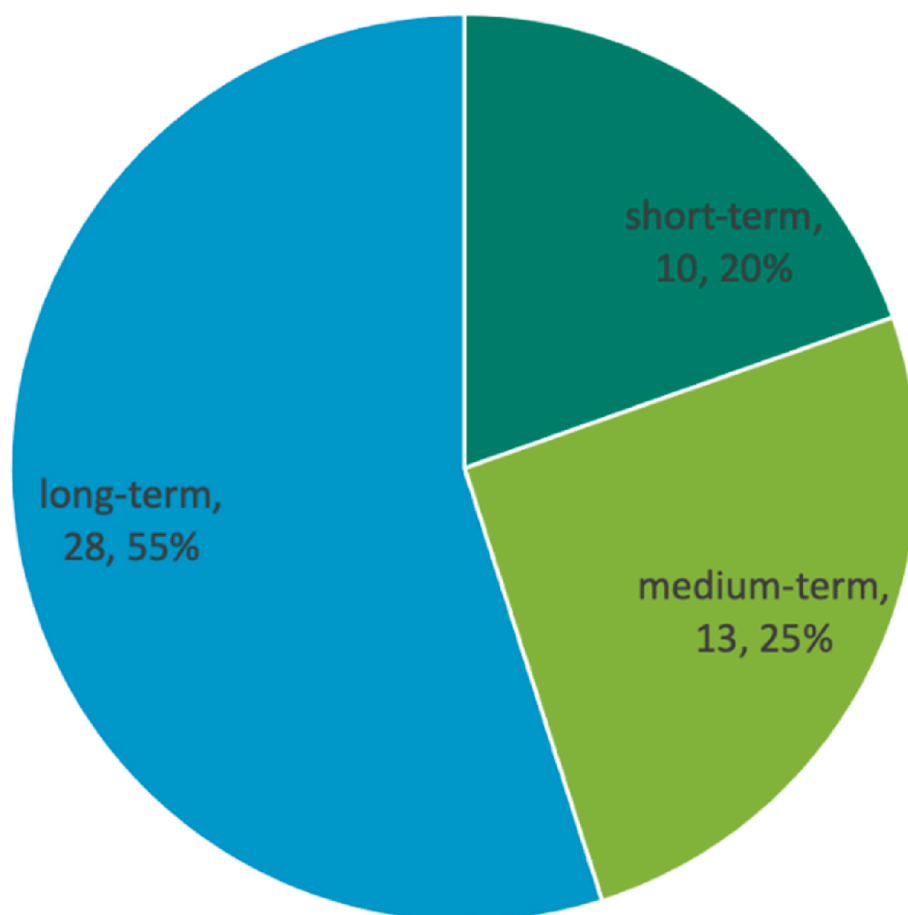


Fig. 3. Study duration analysis.

Distribution of nonclinical repeat-dose toxicity study durations submitted in support of FDA-approved 51 mAb therapeutics between 2020 and 2025 (excluding Tebentafusp, which entirely relied on *in vitro* approaches). Study durations are categorized as ≤ 4 weeks, >4 –13 weeks, and ≥ 26 weeks.

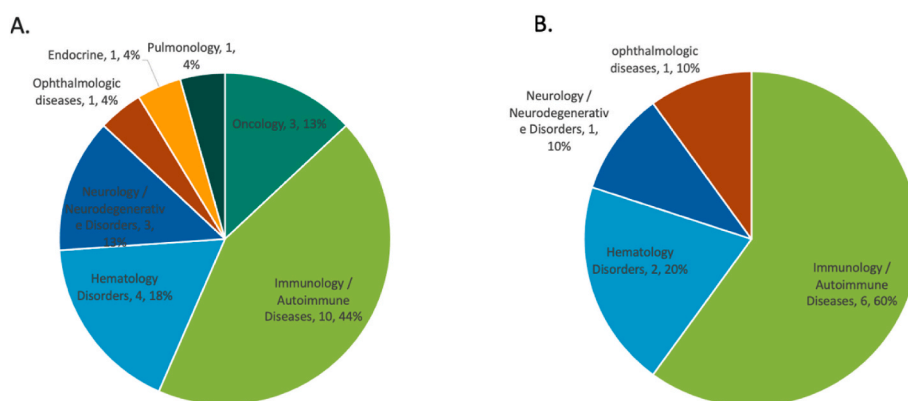


Fig. 4. Therapeutic indications associated with long-term NHP studies.

Therapeutic indication of mAb therapeutics that included both short- or medium-term and long-term NHPs repeat-dose toxicity studies. (A) 23 mAbs with both short- or medium-term and long-term studies. (B) 10 mAbs with both medium-term and long-term studies.

(Fig. 4A and B).

These results indicate that shorter-duration studies are commonly provided in oncology development programs, whereas long-term toxicity studies are more frequently included in non-oncology mAbs nonclinical submissions.

3.2. Species selection and reliance on NHPs

Our analysis of the 52 BLAs included in this study suggests that species selection for mAbs toxicity assessment was driven primarily by tissue cross-reactivity assay. In the dataset of 52 FDA-approved mAbs, NHPs were the most commonly used species due to their pharmacological relevance to human targets: 45 mAbs (86.5%) used cynomolgus monkeys, one Ebola-indicated drug used rhesus monkeys. Six mAbs did

not include NHPs toxicity studies: four drugs lacked any pharmacologically relevant species (two used surrogate murine antibodies, one used transgenic mouse model, one entirely relied on *in vitro* studies), while two demonstrated rodent cross-reactivity to support toxicity assessment in rats or mice.

A total of 35 (67.3%) mAbs relied exclusively on cynomolgus monkeys for the toxicity evaluation due to lack of cross-reactivity in rodents, consistent with ICH S6 (R1) guidance that suggests testing in a single-species when scientifically justified. Among 39 submissions (75%) reporting weak or absent target binding in rodents and identifying cynomolgus monkeys as the only pharmacologically relevant species, six submissions nonetheless conducted rodent toxicity studies enabled by engineered approaches (five used surrogate antibodies, one used transgenic mouse model). Only seven mAbs showed pharmacological activity in rodents, and all of them included toxicity studies in mice or rats (Fig. 5A). Among these, five also included toxicity studies in NHPs, whereas two relied solely on rodent studies for nonclinical toxicity assessment. For five mAbs (teplizumab-mzwv; spesolimab-sbzo; tebentafusp-tebn; inebilizumab-cdon; isatuximab-irfc) no pharmacologically relevant experimental animal species was identified; therefore, nonclinical evaluation relied entirely on surrogate murine antibodies (3/5), transgenic mouse model (1/5), or cynomolgus monkeys to assess potential off-target effects (1/5).

In total, 17 out of 52 mAbs included toxicity studies in species other than NHPs. Among these, 10 (58.8%) of these drugs did not demonstrate pharmacological activity in rodents and relied on engineered approaches: either transgenic mouse models (20%) or surrogate murine antibodies (80%) (Fig. 5B).

Collectively, these observations suggest that while NHPs are often the only scientifically justifiable species for toxicity assessment, multi-species testing strategies are still frequently employed for individual molecules.

3.3. Incremental value of long-term versus short- and medium-term studies

To assess whether long-term NHP toxicity studies provided incremental safety information beyond that obtained from shorter studies, we examined a subset of 23 mAbs that included both short- or medium-term and long-term repeat-dose toxicity studies in cynomolgus monkeys. A summary comparison of short-/medium- and long-term repeat-dose toxicity studies for the 23 mAbs included in this analysis is provided in Table 2.

Findings observed exclusively in long-term studies were classified into four predefined categories which were mentioned in section 2.3, above and illustrated in Fig. 6.

Class 1: No Additional Findings.

Ten mAbs (43.5%) demonstrated no new toxicological findings in ≥26-week studies compared with earlier 4- to 13-week studies. In these

Table 2
Comparison of repeat-dose toxicity studies by duration.

Class	features	examples
Class 1: no additional finding		Lebrikizumab; nemolizumab; pozelimab; tralokinumab; toripalimab; mirikizumab; lecanemab; sutimlimab; evinacumab; Satralizumab
Class 2: pharmacology related finding (increase severity or new risk identified)	Predictable	Concizumab; axatilimab; bimekizumab; mosunetuzumab; tremelimumab
Class 3: immunogenicity-related finding	Predictable Non-human related	Concizumab; rozanolixizumab; ublituximab; faricimab; crovalimab; tremelimumab; Tezepelumab; anifrolumab; eptinezumab teprotumumab
Class 4: additional off-target effect finding		

Comparison of short- or medium-term (4 to 13 weeks) and long-term (≥26 weeks) repeat-dose toxicity studies conducted in cynomolgus monkeys for mAb therapeutics included in the analysis.

cases, long-term studies either confirmed earlier observations or were entirely unremarkable, indicating no additional contribution to toxicity identification beyond shorter-duration studies.

Class 2: Pharmacology related finding (increase severity or new risk identified).

Five mAbs exhibited findings in long-term studies that were mechanistically consistent with their known pharmacological action. These findings included therapeutic effect and non-therapeutic effect, and could be further divided into:

Class 2a: Pharmacology related finding (therapeutic effect).

Example: Concizumab-mtci (BLA 761315).

Concizumab-mtci targets tissue factor pathway inhibitor (TFPI) and was developed for prophylaxis in hemophilia A and B. In repeat-dose studies conducted in cynomolgus monkeys, Concizumab administration resulted in pharmacodynamic effects consistent with its intended mechanism of action, including a procoagulant profile and thrombotic findings observed in multiple tissues (e.g., lung, liver, heart, and lymph nodes).

In short- and medium-term studies, changes in coagulation-related parameters were observed and interpreted in the context of on-target pharmacology. In the 26-week repeat-dose study, these pharmacology-consistent effects were sustained and, in some cases, increased in magnitude with continued dosing, without the emergence of additional, mechanistically distinct toxicities. The overall findings were interpreted as exaggerated on-target pharmacology rather than off-target organ toxicity.

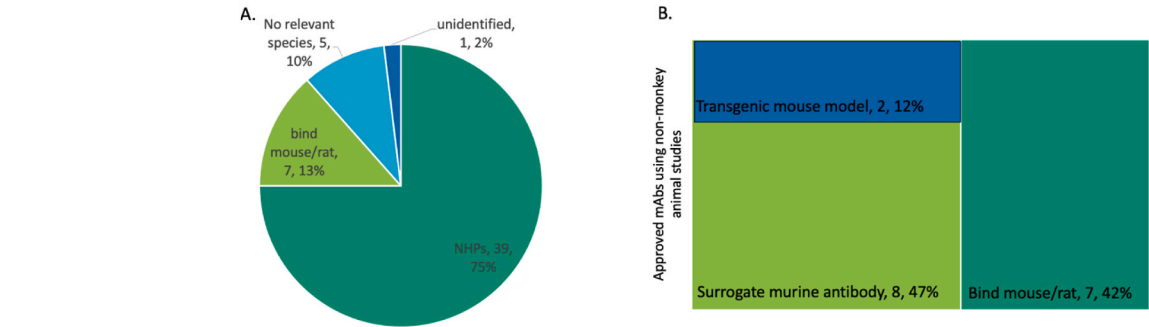


Fig. 5. Species selection for general toxicity assessment. Species used for general toxicity evaluation of FDA-approved mAb therapeutics between 2020 and 2025. (A) Assessment of pharmacological relevance across species. (B) Assessment of pharmacological relevance for models used in rodents.

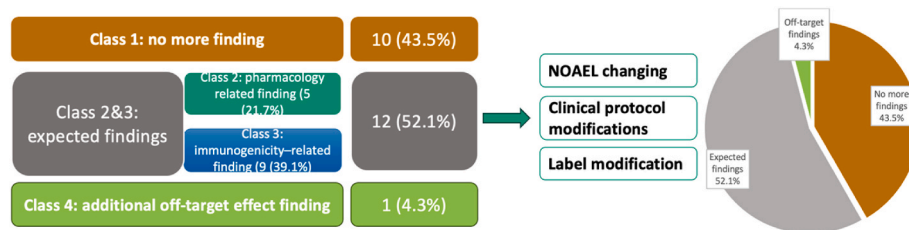


Fig. 6. Comparison of toxicological findings across study durations.

Comparison of toxicological findings identified in short- or medium-term (4 to 13 weeks) versus long-term (≥ 26 weeks) NHP repeat-dose toxicity studies for mAb therapeutics. Findings are categorized based on predefined classification criteria. (Some mAbs showed both pharmacology-related and immunogenicity-related effects in long-term studies.)

These observations were considered in the context of the known mechanism of action and did not alter NOAEL determination, clinical development strategy, or labeling, beyond considerations for pharmacodynamic monitoring.

Class 2b: Pharmacology related finding (non-therapeutic effect).

Example: Axatilimab-csfr (BLA 761411).

Axatilimab targets the colony-stimulating factor 1 receptor (CSF-1R), a key regulator of macrophage survival and function. It is indicated for the treatment of chronic graft-versus-host disease.

In the 13-week toxicity study conducted in cynomolgus monkeys, Axatilimab administration led to periorbital swelling, elevated liver enzymes, and decreased bone biomarkers without any effects on bones at densitometry or histopathology examination. These effects were attributed to the pharmacologic action of CSF-1R inhibition and were not considered adverse. In the 26-week study, additional findings emerged, including dose-dependent accumulation of basophilic material in multiple organs and evidence of bone thickening and growth plate degeneration. These effects were classified as adverse, and as a result, a NOAEL could not be established.

In this case, the long-term study directly impacted regulatory decisions. Bone turnover markers have been incorporated as secondary safety endpoints into clinical trial protocols, and pediatric labeling also includes bone development monitoring. These findings were consistent with exaggerated pharmacological activity rather than off-target toxicity, highlighting a mechanistically expected safety profile.

Class 3: Immunogenicity-related finding.

Nine mAbs demonstrated findings attributed to ADA formation and immune complex deposition. Observations included perivascular inflammation and tissue-specific immune effects. However, these effects were generally species-specific and not relevant to humans. Thus, these findings confounded toxicological interpretation, and generally did not influence NOAEL determination, labeling, or clinical risk management.

Example: Faricimab-svoa (BLA 761235).

Faricimab-svoa (BLA 761235) is a mAb against VEGF-A and Angiopoietin-2, developed for ophthalmic indications.

In short- and medium-term toxicity studies in cynomolgus monkeys, inflammation was observed in ocular and aortic tissues. The sponsor attributed these findings to an ADA response and argued they were not clinically relevant to humans. However, no immunohistochemistry data were provided to support immune complex deposition in the eyes. As concerns regarding the ocular findings remained unresolved, FDA recommended additional immunohistochemical analyses in 26-week studies to differentiate between ADA-related and direct toxic effects. In the long-term study, additional findings emerged, including squamous cysts and plaques in the heart. Although immune complex deposition in the aorta was supported by earlier studies, the 26-week data remained inconclusive regarding whether the findings were caused by direct toxicity or species-specific immune responses. Ultimately, based on a weight-of-evidence approach, including the absence of corresponding human findings and the likelihood of species-specific immunogenicity, the FDA concluded that the heart lesions were of minimal clinical concern.

Class 4: Additional off-target effect findings.

Only one mAb, Teprotumumab, exhibited additional findings in long-term studies. These consisted of mild hematological changes that were later observed in clinical trials but were considered non-adverse and not clinically meaningful, thus did not alter NOAELs, and had no regulatory impact.

Example: Teprotumumab-trbw BLA 761143.

Teprotumumab-trbw (BLA 761143) is an anti-IGF-1R mAb indicated for thyroid eye disease.

In long-term studies, in addition to the consistent results with short-term and medium-term results, including thymus diffuse atrophy and decreased Alkaline Phosphatase (ALP) due to pharmacological activity, mild hematology changes (decreased erythrocyte mass, decreased reticulocytes, decreased neutrophil count) were observed. These hematologic effects were not classified as adverse and did not alter the NOAEL. Notably, similar mild laboratory changes were later reported in clinical trials, supporting the notion that the findings were predictive of pharmacological activity but lacked toxicological or clinical significance. As such, the long-term results had no regulatory impact on labeling or clinical monitoring requirements.

4. Discussion

4.1. Interpretation of the findings

This analysis characterizes whether long-term NHP repeat-dose toxicity studies have substantially contributed to regulatory safety assessment for mAb therapeutics approved by the FDA between 2020 and 2025. Although long-term NHP toxicity studies were included in more than half of the mAb therapeutics development programs reviewed, their incremental contribution to regulatory decision-making was limited. Across the 23 mAbs BLAs evaluated which included both short- or medium-term and long-term NHP studies, only one long-term study influenced regulatory risk management, primarily through enhanced clinical monitoring rather than restriction of clinical development. No long-term study identified an unanticipated toxicity that fundamentally altered the overall regulatory safety assessment, resulted in new clinical contraindications, or led to post-marketing requirements. While within the 23 paired-comparison mAbs, 13/23 (56.5%) of the long-term studies identified findings not observed in shorter-duration studies, the majority of these findings (92.3%) were attributable to predictable pharmacological effects or species-specific immunogenicity rather than novel toxicities. Among all mAbs that included ≥ 26 -week NHP studies, only one case (4.3%) yielded findings that influenced clinical monitoring or risk management measures, and none altered regulatory approval decisions. When additional toxicological findings were observed in long-term studies, they were generally consistent with known target pharmacology or confounded by species-specific immunogenicity and did not result in changes to NOAEL determination, clinical trial design, or labeling. Collectively, these observations indicate that for mAb therapeutics with well-characterized targets and pharmacology, shorter-duration studies generally identified the toxicological

findings relevant to regulatory decision-making.

The findings of this analysis are consistent with prior industry and collaborative evaluations examining the incremental value of long-term NHP toxicity studies for mAbs. Early assessments by Blaich et al. reported that for most mAbs, toxicological profiles observed in six-month studies were comparable to those identified in three-month studies, with indicators of potential toxicity generally detectable in shorter-duration studies (Blaich et al., 2016). Although the early efforts were limited by sample size, they suggested that extended study durations often provided limited additional safety information for mAbs. More recently, larger-scale industry analyses have corroborated these observations. Chien et al. evaluated nonclinical safety packages for 142 mAbs across 11 companies and found that six-month NHP studies rarely contributed additional information beyond that obtained from shorter studies (Chien et al., 2023). This work was conducted as part of a collaborative initiative involving the National Centre for the Replacement, Refinement and Reduction of Animals in Research (NC3Rs), the European Partnership for Alternative Approaches to Animal Testing (EPAA), and the Netherlands Medicines Evaluation Board (MEB), and proposed a weight-of-evidence (WoE) approach to determine the necessity of long-term toxicity studies on a case-by-case basis. Similarly, a meta-analysis by Rana et al. spanning 25 years of 30 biotherapeutic development demonstrated that most clinically relevant toxicities were identified in short-term studies, with long-term studies infrequently influencing human risk assessment (Rana et al., 2024). Together, these prior reports provide important context for the present FDA/CDER/OND analysis, which independently evaluates regulatory review outcomes for recently approved mAbs and demonstrates similar patterns in the limited incremental regulatory impact of long-term NHP studies.

4.2. Relevance to recent FDA guidance on streamlining nonclinical safety studies of mAbs

The findings from this analysis provide descriptive information relevant to the FDA's recently released draft guidance on streamlining nonclinical safety studies for mAbs. Specifically, the results illustrate that routine inclusion of long-term NHP studies has infrequently added additional value to influence regulatory outcomes for those 52 approved mAb therapeutics evaluated in this analysis. Our findings support a pharmacology-driven approach to determining study duration, in which the need for long-term NHP studies is evaluated on a case-by-case basis rather than applied by default. These observations are consistent with the FDA's 2025 Roadmap for Reducing Animal Testing in Preclinical Safety Studies, which emphasizes the use of scientifically justified alternatives, fit-for-purpose study designs, and reduction of redundant animal testing where human relevance is limited. They are also aligned with broader international regulatory and governmental initiatives aimed at reducing unnecessary animal testing, including the EMA Reflection Paper on the use of non-human primates in safety testing and national strategies such as the United Kingdom's roadmap for replacing animal testing (European Medicines Agency, 2025; UK Government, 2025). Importantly, the present analysis does not suggest the elimination of long-term studies in all cases. Rather, it demonstrates that the regulatory value of long-term NHP studies depends on factors such as target biology, therapeutic context, and the presence of unresolved safety signals identified in earlier studies. This present analysis is based on regulatory submissions for mAb therapeutics that progressed to FDA approval, representing development programs considered sufficiently safe to advance through clinical development. This evidence-based perspective is reflected in the current nonclinical safety draft guidance for mAbs, which encourages flexibility in study duration while maintaining appropriate safeguards for patient safety.

4.3. Limitation

Several limitations of this analysis warrant consideration. First, the

dataset was limited to FDA-approved mAb therapeutics and therefore does not include development programs that were discontinued prior to approval. As a result, the analysis does not address whether long-term toxicity findings may have contributed to those mAbs that may have failed during development or those mAb candidates that never made it past discovery and were never submitted to FDA for review. However, the focus on approved products aligns with the objective of evaluating how nonclinical data ultimately inform regulatory decision-making at the time of licensure.

Second, the analysis relied primarily on FDA review documents, which summarize nonclinical findings and their regulatory interpretation but may not capture all details contained in original study reports. Nonetheless, FDA pharmacology/toxicology reviews reflect the findings that FDA reviewers considered relevant to regulatory conclusions, including determinations of NOAELs, clinical risk mitigation strategies, and labeling considerations.

Third, this analysis focused exclusively on mAbs and excluded ADC and other complex biologic modalities. The conclusions should not be generalized to products with distinct mechanisms of action, bio-distribution profiles, or toxicity drivers. In addition, the findings may not apply to mAbs targeting pathways associated with delayed toxicity, developmental biology, or irreversible target engagement, where longer observation periods may be scientifically justified.

Finally, this analysis was not intended to assess the value of long-term studies for exploratory or mechanistic research purposes. Rather, it evaluates their incremental contribution to regulatory safety assessment and decision-making, consistent with the scope of the FDA guidance and the objectives of the present study.

5. Conclusion

This retrospective analysis of FDA-approved mAbs describes the limited contribution of long-term NHP repeat-dose toxicity studies to the prediction of unexpected safety risks in the development program of mAb therapeutics. Across the mAbs evaluated, long-term studies infrequently identified findings that altered regulatory conclusions regarding human risk. When additional observations were noted, they were largely consistent with known target pharmacology or reflected species-specific immunogenicity and generally did not influence NOAEL determination, clinical development strategies, or labeling outcomes.

These findings illustrate that, for mAb therapeutics with well-characterized targets and pharmacology, shorter-duration toxicity studies in biologically relevant species, typically identified the toxicological findings that support regulatory decision-making. At the same time, the analysis underscores that the regulatory value of long-term studies is context-dependent and may vary based on factors such as target biology, therapeutic indication, and unresolved safety signals identified in earlier studies. In this context, the continued development and application of validated NAMs, particularly those capable of assessing target organ-specific toxicities, present an opportunity to complement conventional nonclinical studies by providing mechanistically relevant, human-focused information that could improve the identification and interpretation of potential safety risks in mAb therapeutics development.

Together, these data provide descriptive information and analysis into recent FDA review experience and support the current draft guidance encouraging a flexible, pharmacology-driven approach to nonclinical study duration while maintaining appropriate safeguards for patient safety.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT (OpenAI) to assist with language editing and formatting. After using this tool, the authors reviewed and edited the content as needed and take full

responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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